

## IFN647 Week 8 Review Questions

**Question 1** Which of the following descriptions is wrong? and justify your answer.

- (1) In the query likelihood retrieval model, documents are ranked by the probability that the query text could be generated by the document language model.
- (2) To use the unigram language model to estimate  $P(Q|D)$  - the document  $D$ 's score for the given query  $Q = \{q_1, q_2, \dots, q_n\}$ , the language model probabilities  $P(q_i|D)$  ( $i=1, 2, \dots, n$ ) are needed to be estimated.
- (3) In contrast to the query likelihood model, the document likelihood model  $P(D|Q)$  is more theoretical in the context of information filtering, because there is no explicit query available.
- (4) Kullback-Leibler (KL) divergence is a technique for avoiding this estimation problem and overcoming data sparsity, which means that we typically do not have large amounts of text to use for the language model probability estimates.

### Question 2 (Pseudo relevance feedback)

Assume your user provides a query  $Q = \{q_1, q_2, \dots, q_m\}$  and a data collection  $C = \{d_1, d_2, \dots, d_N\}$  (unlabelled dataset) and asks you to define a probabilistic IF model by using pseudo relevance feedback techniques. After your data exploration of  $C$ , and you find that basically the distribution of terms in relevant documents is independent and their distribution in non-relevant documents is independent. And you want to define the probable relevance based only on the presence of search terms in documents. Describe your thoughts on how to design this IF model.

**Tip:** Review the Pseudo-Feedback Algorithm in the lecture notes.

### Question 3 (Pseudo-Relevance Feedback Style CoT Prompt)

The following is a prompting template for query expansion using pseudo-relevance feedback.

**Task:** Expand the query using pseudo-relevance feedback.  
**Query:** [Original query]  
**Top documents:** [Short snippets]  
**Instructions:** Briefly explain your reasoning in **2-3 steps**, then provide the expanded query.

Design a Pseudo-Relevance Feedback-style Chain-of-Thought (CoT) prompt for the original query “climate change impact”, and test it in ChatGPT.

**Tip:** You may use Google News search to collect a few short snippets. Use these snippets to compare the search results obtained with the **original query** versus the **expanded query**.

**Question 4.** Assume  $C$  is a collection of tweets. We can use the following query likelihood method to calculate the probability of a tweet  $d$  occurring for a given query  $Q$ .

$$P(Q, d) = \sum_{w \in Q, d} c(w, Q) \log \left( 1 + \frac{c(w, d)}{\mu \times P(w|C)} \right) + |Q| \log \frac{\mu}{\mu + |d|}$$

Let  $|d|$  and  $|Q|$  are the respective lengths (size) of the tweet  $d$  and query  $Q$ , and  $\mu$  is the smoothing parameter. Interpret the meaning of  $c(w, d)$ ,  $c(w, Q)$ , and  $P(w|C)$ .